

B21 — Cybersecurity Learning Activity Design

Phishing Detection for University Students

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Title	Can You Spot the Phish? — A Phishing Detection Workshop
Target Audience	First or second-year university students with no formal cybersecurity background. The learner uses email and online accounts daily but has limited awareness of social engineering tactics. They are comfortable with smartphones and basic web browsing.
Duration	15 minutes (suitable for a single sitting between study sessions)
Materials Required	• Facilitator: 3 printed or on-screen phishing email examples (1 obvious, 1 moderate, 1 subtle) • Learner: pen and paper (or phone to type answers) • 5-question pre/post quiz (printed or shared as a note) • Optional: laptop to show live browser URL inspection

Learning Objectives

By the end of this activity, the learner will be able to:

- **LO1** — Identify at least 3 common indicators of a phishing email (mismatched sender address, urgency language, suspicious links).
- **LO2** — Correctly classify a given email sample as legitimate or phishing with reasoning.
- **LO3** — Explain one safe action to take upon receiving a suspected phishing email.
- **LO4** — Describe why phishing is an effective attack vector against non-technical users.

Activity Structure

Time	Phase	Activity
0–2 min	Introduction	Ask the learner 3 warm-up questions (pre-quiz): What is phishing? How do you make a strong password? What is 2FA? Note their answers — this is baseline evidence.

2–5 min	Core Concept	Explain phishing in plain language using the 'disguised letter' analogy. Introduce the 3 key red flags: sender mismatch, urgency/fear language, suspicious hyperlinks.
5–11 min	Worked Examples	Walk through 3 real-style email examples together. For each: ask the learner what they notice first, then reveal the red flags one by one. Use the laptop to demonstrate hovering over a link to reveal the true URL.
11–14 min	Practice Quiz	Learner independently answers the 5-question post-quiz. Facilitator reviews answers aloud and explains any mistakes.
14–15 min	Debrief	Ask the learner what surprised them most. Ask them to name one thing they will do differently with their email from now on. Take a photo of the session as evidence.

Assessment

Learning is assessed at two points:

Point	Method	Success Indicator
Pre-session (baseline)	3 verbal warm-up questions; answers noted by facilitator	Establishes prior knowledge level
Post-session (outcome)	5-question written quiz (classify emails + name red flags)	Score of 4/5 or above; correct identification of red flags

Additionally, the learner is asked to send a confirmation text message after the session (e.g. 'Thanks for the phishing lesson — I had no idea about the sender address trick!'). This message is screenshot as supplementary evidence for B14.

Pedagogical Rationale

This activity applies three evidence-based instructional design principles:

- **Scenario-based learning** — Scenario-based learning: Real-style phishing email examples ground abstract concepts in lived experience. Research by Kumaraguru et al. (2007) demonstrated that embedded, example-driven training produces significantly better phishing detection rates than policy-only awareness campaigns. Rather than

telling the learner 'phishing is dangerous,' the activity places them in the role of a detective evaluating actual artefacts.

- **Pre/post assessment** — Pre/post assessment for metacognitive awareness: By explicitly asking the learner what they knew before and what they know now, the design encourages reflection on the learning process itself. This is consistent with Bloom's Taxonomy — the debrief question ('what will you do differently?') targets the Apply and Evaluate levels rather than simple recall.
- **Minimal effective dose** — Minimal effective dose: At 15 minutes, the session is deliberately brief. Attention research (Bligh, 2000) suggests that information retention drops sharply after 10–15 minutes in a non-interactive format. By limiting scope to three core red flags and reinforcing them immediately with examples and a quiz, the design prioritises depth over breadth — a key principle for security awareness training aimed at non-specialists.

Phishing was selected as the topic because it represents the single most common initial access vector in real-world cybersecurity incidents (Verizon DBIR, 2023), making it the highest-impact topic for a general audience. The skills transfer directly — a learner who can classify a phishing email in a 15-minute session is genuinely more secure the next day they check their inbox.

GitHub Evidence

This document is uploaded to the Portfolio GitHub repository at:

[part-2/b21-learning-activity/b21-activity-design.docx](#)

- **b21-activity-design.docx** — Activity Design Document (this file)
- **b21-phishing-quiz.docx** — 5-question phishing detection quiz used in the session
- **b21-session-photo.jpg** — Photo of the teaching session (taken during B14 delivery)
- **b21-confirmation-text.png** — Screenshot of confirmation text from the learner